SWLG: An Intro to Single Event Upsets

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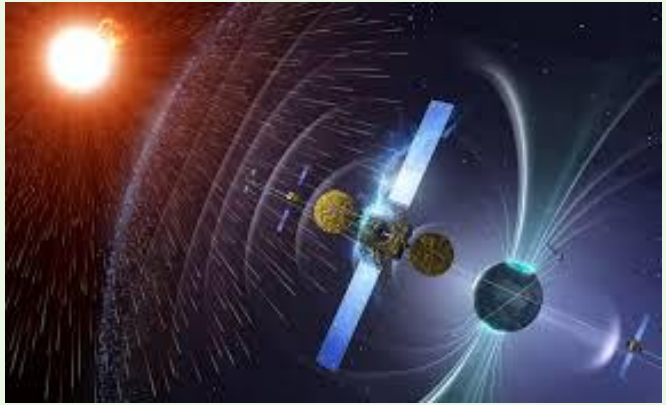
Topics



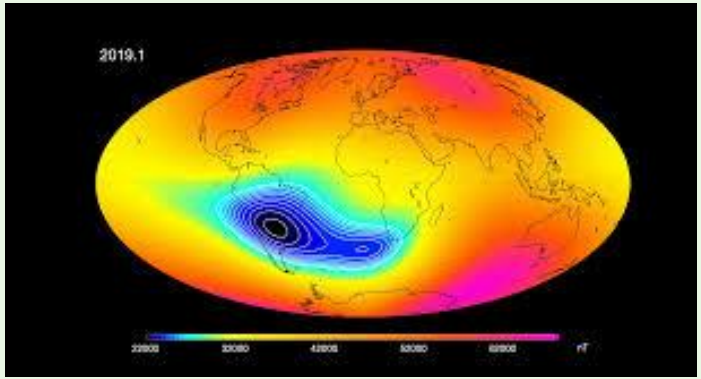
1. **Single Event Upsets (SEU)**
- **What are they?**

2. **Related Works and Datasets, How do we predict?**
- Paper: A Machine Learning Approach to Predicting SEP Proton Intensity and Events Using Time Series of Relativistic Electron Measurements **- AGU**
  - Dataset: GOES Proton Flux DB
- Paper: A Machine Learning Approach to Predicting Single Event Upsets **- Nanyang Technological University, Singapore**
  - Dataset: "Flying Laptop"
- Paper: Single Event Phenomena: Testing and Prediction - **JHU**
  - Dataset: CRRES PROTEL 1-min Proton Distribution Functions

3. **Other Datasets**

Single Event Upsets: What Are They?

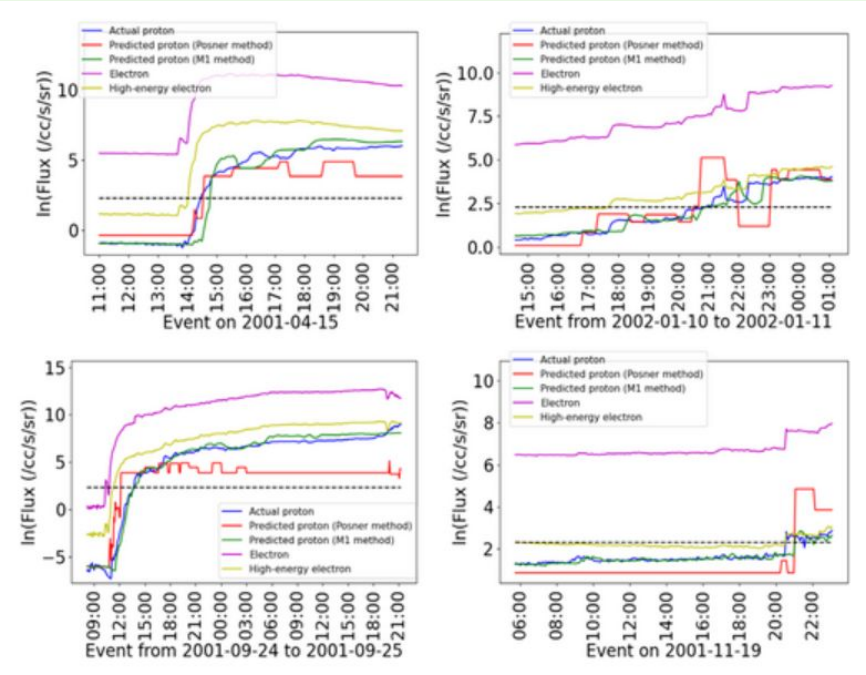
- A Single Event Upset (SEU) are soft errors on semiconductor devices on space platforms due to exposure to Solar Energetic Particles (SEP) from the surrounding space environment
  - SEU Energy Levels between keV to a few MeV
  - Soft Errors: Bit flips in memory units that do no permanently destruct the functions of a device: SRAM, RAM, Junction Areas
    - occurred when the charged deposited energy from a colliding proton exceeds a device specific critical charge
  - SEPs includes protons, alpha particles, heavy nucleons, and electrons. However in the case for SEU, high energy protons are the biggest concern
  - The environment usually due to the Van Allen Belts as almost every LEO SAT has experienced SEU. Solar activity from solar winds as well as galactic/cosmic microwave background radiation.



South Atlantic Anomaly

Single Event Upsets: How do we predict?

- AGU study seeks to develop an advanced warning system of SEUs using neural networks with data fed by GOES satellites measuring proton fluxes to predict future proton flux intensity
  - Define three phases of SEU: rising, falling, background -> in reference to peak flux > 10 MeV
  - Determine Error and Lag between actual proton and predicted proton with mean absolute error
    - Minimize MAE
  - Compared values using three neural network models vs just one (M3, M1) and Phases vs no phases, NN vs RNN



Single Event Upsets: GOES Dataset

GOES data values all have an associated quality flag to decode with quality flag table.

Dataset spans: 1974 - 2020 where GOES 13-15 have been discontinued

Data format: .csv, .netcdf

Data included in each file has ephemeral data of satellite as well as average number of counts per second and average flux of electrons from different onboard detectors. Each field has an associated quality flag boolean value (0,1)

```
satellite location:
time_tag_orbit,west_longitude,inclination
2020-02-19 00:00:00,128.4, 0.371

data:
time_tag,E2E_QUAL_FLAG,E2E_UNCOR_CR,E2E_UNCOR_FLUX,E2W_QUAL_FLAG,E2W_UNCOR_CR,E2W_UNCOR_FLUX
2020-02-19 00:00:14.177,0,1.1585E+00,2.3171E+01,0,2.1037E+00,4.2073E+01
2020-02-19 00:00:30.563,0,1.0366E+00,2.0732E+01,0,1.4634E+00,2.9268E+01
```

Single Event Upsets: Flying Laptop

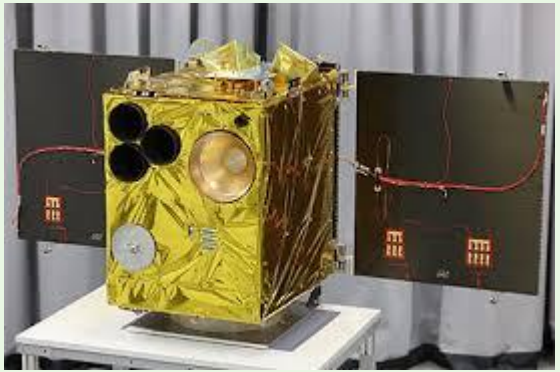
“Flying Laptop satellite have provided insights both on the in orbit performance of SRAM technology in LEO as well as the space radiation environment of the declining Solar Cycle 24

EDAC system detects errors in SRAM memory, counts as detection

Data available here at [Stuttgart University]

TLE of flying laptop over time and detected SEU events with time tags to map on TLE Data

.csv file format



Single Event Upsets: Flying Laptop

[Nanyang Technological University]

CREMER NN Model -> need to find model

They had data for when SEU occurred but needed generated data for negative cases

“CREMER uses a soft voting ensemble to combine extreme gradient boosting and random forests. “

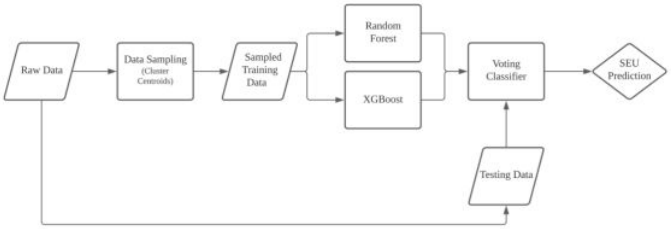


Figure 3: Model Training Pipeline

Single Event Upsets: Prediction Methods - JHU

- Linear Energy Transfer (LET) is the deposition of energy by high energy protons onto the cells of a semiconductor device as it pierces through the device

$$L_{eff} = \frac{L}{cos\theta}$$

- Path- Length Distribution Model analyzes the frequency of SUE's striking onboard memory cells,

$$N = 22.5\pi A Q_C \int_0^\infty \frac{D[p(L)]}{L^2} \psi(L) dL$$

- Bendel Proton Upset Model provides an estimate of the upset rate due to proton nuclear reactions, secondary emissions

$$\sigma = (24/A)^{14}[1 - exp(-0.18Y^{1/2})]^4$$
$$Y = (18/A)^{1/2}(E - A)$$

Single Event Upsets: CRRES Dataset

CRRES Dataset from the Proton Telescope (PROTEL) instrument on the CRRES satellite includes equatorial pitch angle distributions of one-minute flux averages for twenty-four proton energy channels at 1 to 100 MeV.

Provided in UNIX binary format along with FORTRAN software for extraction of selected parameters

Files are .pad, Act! database-> investigating how to open and use. Microsoft has a few applications to choose from

Single Event Upsets: Other Databases

All other databases located at Space Physics Data Facility

- Advanced Composition Explorer (ACE)
  - Electron Proton Alpha particle Monitor (EPAM) .csv
  - SWEPAM Solar wind Proton Alpha Monitor .csv
- Absolute Radiometer for Cosmology Astrophysics, and Diffuse Emissions (ARCAD)
  - Particle Flux data
- HELIOS 1&2
  - Measured electrons in three ranges between 2 and 8 MeV, and protons and alpha particles in three ranges between 20 and 56 MeV/n
- Pioneer 10/11
  - Particle Flux data, lots of other things no label
  - https://theses.hal.science/tel-01954572/file/Viyas_Gupta_2017.pdf
- Voyager 1&2
  - Cosmic Ray fluxes and Low energy Charged Particle
  - Voyager affected by radiation
- WIND
  - Measures solar wind fluxes onto itself
- GPS
  - https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024SW004042